# ■ TIMB BANK

## Enterprise Network Implementation

*Complete Project Documentation*

| | |
|---|---|
| **Student** | Jessica Ugoh |
| **Institution** | Nigeria |
| **Program** | Electronics and Computer Engineering |
| **Tool Used** | Cisco Packet Tracer |
| **Project Type** | Enterprise Bank Network Simulation |
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| | | | |
|---|---|---|---|
| **1 HQ<br>+ 3 Branches** | **10 VLANs<br>per site** | **Full Mesh<br>Frame Relay** | **ASA<br>Firewall** |
| **VoIP<br>Inter-site** | **GRE VPN<br>Tunnels** | **RADIUS<br>AAA** | **Online<br>Banking** |

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# 1. PROJECT OVERVIEW

This document presents the complete implementation of the TIMB Bank enterprise network simulation using Cisco Packet Tracer. The project simulates a real-world banking network infrastructure for TIMB Bank (Trusted. Innovative. Modern. Banking.) with one headquarters and three branch offices.

## Project Objectives

The TIMB Bank network was designed to achieve the following objectives:

| Objective | Description | Status |
|---|---|---|
| Connectivity | All sites communicate seamlessly | ■ Done |
| Security | Bank-grade firewall and access control | ■ Done |
| Redundancy | No single point of failure for internet | ■ Done |
| Voice | VoIP calling between all sites | ■ Done |
| Management | Centralized monitoring and logging | ■ Done |
| Authentication | RADIUS-based access control | ■ Done |
| Services | Web, Email, Online Banking | ■ Done |
| Backup | FTP configuration backup | ■ Done |

## Technologies Used

| Technology | Purpose | Standard |
|---|---|---|
| Cisco IOS 15.1 | Router/Switch OS | Industry Standard |
| OSPF | Dynamic routing protocol | RFC 2328 |
| Frame Relay | WAN connectivity | ITU-T I.233 |
| GRE Tunnels | Site-to-site VPN | RFC 2784 |
| 802.1Q VLANs | Network segmentation | IEEE 802.1Q |
| ASA Firewall | Perimeter security | Cisco ASA 5505 |
| NAT/PAT | Address translation | RFC 3022 |
| SSH v2 | Secure management | RFC 4253 |
| RADIUS/AAA | Authentication | RFC 2865 |
| NTP | Time synchronization | RFC 5905 |
| Syslog | Event logging | RFC 5424 |
| VoIP/CME | Voice communication | SIP/H.323 |
| DHCP | Auto IP assignment | RFC 2131 |
| SMTP/POP3 | Email services | RFC 5321 |

# 2. NETWORK TOPOLOGY

## Topology Overview

The TIMB Bank network uses a Full Mesh WAN topology connecting HQ and all three branches via Frame Relay. Each site has its own Multilayer Switch (MLS) for inter-VLAN routing, access switches for end devices, and a CME Router for VoIP services.

```
INTERNET (8.8.8.x)
|
ISP Router (203.0.113.1)
|
ASA Firewall (203.0.113.2 | 192.168.1.1 | 172.16.0.1)
| |
DMZ Servers HQ Router (192.168.1.2)
172.16.0.10-12 |
HQ-MLS (172.31.0.2)
/ | \
VLAN VLAN VLAN VLAN
GRP1 GRP2 GRP3 GRP4

Frame Relay Full Mesh WAN:
HQ ←——→ Branch1 (10.0.0.0/30)
HQ ←——→ Branch2 (10.0.1.0/30)
HQ ←——→ Branch3 (10.0.2.0/30)
B1 ←——→ B2 (10.1.0.0/30)
B1 ←——→ B3 (10.2.0.0/30)
B2 ←——→ B3 (10.3.0.0/30)
```

## Site Structure

| Site | Router | MLS | Access Switches | Servers | CME Router |
|---|---|---|---|---|---|
| HQ | HQ-Router | HQ-MLS | 4 x 2960 | 10 servers | 192.168.70.2 |
| Branch 1 | B1-Router | B1-MLS | 4 x 2960 | 4 servers | 172.20.70.2 |
| Branch 2 | B2-Router | B2-MLS | 4 x 2960 | 4 servers | 172.21.70.2 |
| Branch 3 | B3-Router | B3-MLS | 4 x 2960 | 4 servers | 172.22.70.2 |

# 3. IP ADDRESSING SCHEME

## HQ VLANs — 192.168.x.x

| VLAN | Name | Network | Gateway | DHCP Helper |
|---|---|---|---|---|
| 10 | ADMIN-MANAGEMENT | 192.168.10.0/24 | 192.168.10.1 | 192.168.200.10 |
| 20 | CORE-BANKING | 192.168.20.0/24 | 192.168.20.1 | 192.168.200.10 |
| 30 | CLEARING-SWIFT | 192.168.30.0/24 | 192.168.30.1 | 192.168.200.10 |
| 40 | TELLERS | 192.168.40.0/24 | 192.168.40.1 | 192.168.200.10 |
| 50 | CUSTOMER-SERVICE | 192.168.50.0/24 | 192.168.50.1 | 192.168.200.10 |
| 60 | CCTV-SECURITY | 192.168.60.0/24 | 192.168.60.1 | 192.168.200.10 |
| 70 | VOIP-PHONES | 192.168.70.0/24 | 192.168.70.2 | CME Router |
| 80 | ATMs | 192.168.80.0/24 | 192.168.80.1 | 192.168.200.10 |
| 90 | GUEST-WIFI | 192.168.90.0/24 | 192.168.90.1 | 192.168.200.10 |
| 100 | IT-MANAGEMENT | 192.168.100.0/24 | 192.168.100.1 | 192.168.200.10 |
| 200 | HQ-SERVERS | 192.168.200.0/24 | 192.168.200.1 | — |

## Branch IP Ranges

| Branch | VLAN Range | Server VLAN | Router LAN | MLS Uplink |
|---|---|---|---|---|
| Branch 1 | 172.20.x.x | 172.20.200.0/24 | 172.20.0.1/30 | 172.20.0.2/30 |
| Branch 2 | 172.21.x.x | 172.21.200.0/24 | 172.21.0.1/30 | 172.21.0.2/30 |
| Branch 3 | 172.22.x.x | 172.22.200.0/24 | 172.22.0.1/30 | 172.22.0.2/30 |

## WAN Links — Frame Relay Full Mesh

| Link | HQ Side | Branch Side | HQ DLCI | Branch DLCI |
|---|---|---|---|---|
| HQ ↔ Branch1 | 10.0.0.1/30 | 10.0.0.2/30 | 102 | 201 |
| HQ ↔ Branch2 | 10.0.1.1/30 | 10.0.1.2/30 | 103 | 301 |
| HQ ↔ Branch3 | 10.0.2.1/30 | 10.0.2.2/30 | 104 | 401 |
| B1 ↔ B2 | 10.1.0.1/30 | 10.1.0.2/30 | 123 | 213 |
| B1 ↔ B3 | 10.2.0.1/30 | 10.2.0.2/30 | 132 | 312 |
| B2 ↔ B3 | 10.3.0.1/30 | 10.3.0.2/30 | 231 | 321 |

## DMZ Servers

| Server | IP Address | Services | Public IP (NAT) |
|---|---|---|---|
| Web Server | 172.16.0.10 | HTTP, HTTPS, DNS | 203.0.113.10 |
| Email Server | 172.16.0.11 | SMTP, HTTP | 203.0.113.11 |
| Online Banking | 172.16.0.12 | HTTP, HTTPS | 203.0.113.12 |

## HQ Servers — 192.168.200.x

| IP | Server Name | Services |
|---|---|---|
| 192.168.200.10 | DHCP-SERVER | DHCP — 10 pools for VLANs 10-100 |
| 192.168.200.11 | DNS-SERVER | DNS — A records for timb.com domains |
| 192.168.200.12 | RADIUS-SERVER | AAA authentication |
| 192.168.200.13 | NTP-SERVER | Network Time Protocol |
| 192.168.200.14 | SYSLOG-SERVER | Centralized event logging |
| 192.168.200.15 | DATABASE-SERVER | HTTP/HTTPS |
| 192.168.200.16 | BACKUP-SERVER | FTP config backup |
| 192.168.200.17 | EMAIL-SERVER | SMTP/POP3 internal email |
| 192.168.200.18 | FILE-SERVER | FTP file sharing |
| 192.168.200.19 | MONITORING-SERVER | HTTP/Syslog monitoring |

# 4. VLAN CONFIGURATION

## HQ Switch Groups

HQ uses 4 VLAN group access switches connected to the HQ-MLS via trunk ports. Each switch group serves specific departments.

| Switch | VLANs | Departments |
|---|---|---|
| VLAN-GROUP1 | 10, 20 | Admin Management, Core Banking |
| VLAN-GROUP2 | 30 | Clearing/SWIFT |
| VLAN-GROUP3 | 50, 60 | Customer Service, CCTV Security |
| VLAN-GROUP4 | 40, 70, 80, 90, 100 | Tellers, VoIP, ATMs, Guest WiFi, IT |
| Server-Switch | 200 | HQ Servers |

## Key VLAN Commands — HQ MLS

```
vlan 10
name ADMIN-MANAGEMENT
vlan 20
name CORE-BANKING
vlan 30
name CLEARING-SWIFT
vlan 40
name TELLERS
vlan 50
name CUSTOMER-SERVICE
vlan 60
name CCTV-SECURITY
vlan 70
name VOIP-PHONES
vlan 80
name ATMs
vlan 90
name GUEST-WIFI
vlan 100
name IT-MANAGEMENT
vlan 200
name HQ-SERVERS
```

## Guest VLAN Restriction — ACL

A special ACL was configured on VLAN 90 (Guest WiFi) to prevent guests from accessing internal bank resources while allowing internet access.

```
ip access-list extended GUEST-RESTRICT
permit tcp 192.168.90.0 0.0.0.255 any eq www
permit tcp 192.168.90.0 0.0.0.255 any eq 443
permit udp 192.168.90.0 0.0.0.255 any eq 53
deny ip 192.168.90.0 0.0.0.255 192.168.0.0 0.0.255.255
permit ip any any
```

```
interface Vlan90
ip access-group GUEST-RESTRICT in
```

# 5. ROUTING — OSPF

## OSPF Design

| Device | Router ID | Networks Advertised |
| --- | --- | --- |
| HQ-Router | 1.1.1.1 | WAN links, HQ uplink |
| B1-Router | 2.2.2.2 | WAN links, B1 LAN |
| B2-Router | 3.3.3.3 | WAN links, B2 LAN |
| B3-Router | 4.4.4.4 | WAN links, B3 LAN |
| HQ-MLS | 5.5.5.5 | All HQ VLANs (10-100, 200) |
| B1-MLS | 6.6.6.6 | All B1 VLANs |
| B2-MLS | 7.7.7.7 | All B2 VLANs |
| B3-MLS | 8.8.8.8 | All B3 VLANs |

## HQ Router OSPF Config

```
router ospf 1
router-id 1.1.1.1
network 10.0.0.0 0.0.0.3 area 0
network 10.0.1.0 0.0.0.3 area 0
network 10.0.2.0 0.0.0.3 area 0
network 172.31.0.0 0.0.0.3 area 0
network 192.168.1.0 0.0.0.3 area 0
```

# 6. WAN — FRAME RELAY FULL MESH

## Why Full Mesh?

A full mesh topology was chosen to ensure every site can communicate directly with every other site without traffic having to traverse HQ. This simulates an E-LAN Metro Ethernet service providing maximum redundancy and performance.

## HQ Router Frame Relay Config

```
interface Serial0/0/0
encapsulation frame-relay

interface Serial0/0/0.1 point-to-point
ip address 10.0.0.1 255.255.255.252
frame-relay interface-dlci 102

interface Serial0/0/0.2 point-to-point
ip address 10.0.1.1 255.255.255.252
frame-relay interface-dlci 103

interface Serial0/0/0.3 point-to-point
ip address 10.0.2.1 255.255.255.252
frame-relay interface-dlci 104
```

## Metro Ethernet Service Type

The TIMB Bank network implements E-LAN (Ethernet LAN) Metro Ethernet service using full mesh Frame Relay — every site communicates directly with every other site. In production, this would be replaced with MPLS VPN with SD-WAN management.

# 7. ASA FIREWALL & DMZ

## ASA Interface Design

| Interface | Name | IP Address | Security Level | Purpose |
|---|---|---|---|---|
| GigabitEthernet1/1 | outside | 203.0.113.2/30 | 0 | Internet facing |
| GigabitEthernet1/2 | inside | 192.168.1.1/30 | 100 | Internal network |
| GigabitEthernet1/3 | dmz | 172.16.0.1/24 | 50 | DMZ servers |

## Security Level Policy

ASA security levels automatically control traffic flow without ACLs:

```
Inside (100) → can reach DMZ (50) and Outside (0) ■
DMZ (50) → can reach Outside (0) only ■
Outside (0) → cannot reach Inside or DMZ ■ (blocked by default)
```

## Traffic Policy

| Direction | Traffic | Action |
|---|---|---|
| LAN → Internet | TCP 80, 443, UDP 53, ICMP | PERMIT |
| LAN → DMZ | TCP 80, 443, ICMP | PERMIT |
| Internet → DMZ | TCP 80, 443 to servers only | PERMIT |
| Internet → DMZ | ICMP | DENY |
| DMZ → Internet | TCP 80, 443, 53 | PERMIT |
| DMZ → LAN | All traffic | DENY |
| Internet → LAN | All traffic | DENY |

# 8. NAT CONFIGURATION

## NAT Objects

| Object | Network | NAT Type | Translated To |
|---|---|---|---|
| HQ_NET | 192.168.0.0/16 | Dynamic PAT | 203.0.113.2 (ASA outside) |
| BRANCH1_NET | 172.20.0.0/16 | Dynamic PAT | 203.0.113.2 (ASA outside) |
| BRANCH2_NET | 172.21.0.0/16 | Dynamic PAT | 203.0.113.2 (ASA outside) |
| BRANCH3_NET | 172.22.0.0/16 | Dynamic PAT | 203.0.113.2 (ASA outside) |
| WEB-SERVER | 172.16.0.10 | Static NAT | 203.0.113.10 |
| EMAIL-SERVER | 172.16.0.11 | Static NAT | 203.0.113.11 |
| BANKING-SERVER | 172.16.0.12 | Static NAT | 203.0.113.12 |

## Branch Direct Internet NAT

Each branch also has its own NAT for direct internet access:

```
! Branch 1 Router
ip access-list standard BRANCH1-NAT
permit 172.20.0.0 0.0.255.255
ip nat inside source list BRANCH1-NAT interface GigabitEthernet0/1 overload

! ISP Router VLANs for branch links
interface Vlan201
ip address 100.0.1.1 255.255.255.252 ← Branch 1 internet link
interface Vlan202
ip address 100.0.2.1 255.255.255.252 ← Branch 2 internet link
interface Vlan203
ip address 100.0.3.1 255.255.255.252 ← Branch 3 internet link
```

# 9. INTERNET CONNECTIVITY

## Two-Path Internet Architecture

TIMB Bank implements a redundant internet architecture where both HQ and each branch have independent internet paths. This ensures that if HQ internet fails, all branches remain online.

| Site | Internet Path | ISP Link | Status |
|---|---|---|---|
| HQ | Via ASA Firewall | 203.0.113.0/30 | ■ Active |
| Branch 1 | Direct to ISP | 100.0.1.0/30 (VLAN 201) | ■ Active |
| Branch 2 | Direct to ISP | 100.0.2.0/30 (VLAN 202) | ■ Active |
| Branch 3 | Direct to ISP | 100.0.3.0/30 (VLAN 203) | ■ Active |

## Internet Servers

| Server | IP | Services |
|---|---|---|
| DNS Server | 8.8.8.8 | DNS resolution for internet domains |
| Google Server | 8.8.8.2 | Simulates Google web service |

# 10. VOIP INTER-SITE CALLING

## CME Router Design

| Site | CME Router IP | Extensions | Dial-peers to |
|---|---|---|---|
| HQ | 192.168.70.2 | 1001-1020 | B1(2xxx), B2(3xxx), B3(4xxx) |
| Branch 1 | 172.20.70.2 | 2001-2020 | HQ(1xxx), B2(3xxx), B3(4xxx) |
| Branch 2 | 172.21.70.2 | 3001-3020 | HQ(1xxx), B1(2xxx), B3(4xxx) |
| Branch 3 | 172.22.70.2 | 4001-4020 | HQ(1xxx), B1(2xxx), B2(3xxx) |

## HQ CME Dial-peer Config

```
telephony-service
max-ephones 10
max-dn 10
ip source-address 192.168.70.2 port 2000
auto assign 1 to 10

! Dial-peer to Branch 1
dial-peer voice 21 voip
destination-pattern 2...
session target ipv4:172.20.70.2
codec g711ulaw

! Dial-peer to Branch 2
dial-peer voice 31 voip
destination-pattern 3...
session target ipv4:172.21.70.2
codec g711ulaw

! Dial-peer to Branch 3
dial-peer voice 41 voip
destination-pattern 4...
session target ipv4:172.22.70.2
codec g711ulaw
```

# 11. GRE VPN TUNNELS

## Why GRE Tunnels?

IPSec VPN could not be configured due to Packet Tracer IOS limitations — the security license was not available. GRE (Generic Routing Encapsulation) tunnels were used instead to create virtual private links between sites. In production, IPSec encryption would be added over the GRE tunnels.

| Tunnel | HQ IP | Branch IP | Source | Destination |
|---|---|---|---|---|
| Tunnel1 (HQ-B1) | 10.10.1.1/30 | 10.10.1.2/30 | Se0/0/0.1 | 10.0.0.2 |
| Tunnel2 (HQ-B2) | 10.10.2.1/30 | 10.10.2.2/30 | Se0/0/0.2 | 10.0.1.2 |
| Tunnel3 (HQ-B3) | 10.10.3.1/30 | 10.10.3.2/30 | Se0/0/0.3 | 10.0.2.2 |

## HQ Router GRE Config

```
interface Tunnel1
ip address 10.10.1.1 255.255.255.252
tunnel source Serial0/0/0.1
tunnel destination 10.0.0.2
no shutdown

interface Tunnel2
ip address 10.10.2.1 255.255.255.252
tunnel source Serial0/0/0.2
tunnel destination 10.0.1.2
no shutdown

interface Tunnel3
ip address 10.10.3.1 255.255.255.252
tunnel source Serial0/0/0.3
tunnel destination 10.0.2.2
no shutdown
```

# 12. SECURITY FEATURES

## SSH Version 2

All routers and switches are secured with SSH v2 — Telnet is disabled.

```
ip domain-name timb.com
crypto key generate rsa ! Key size: 1024
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
username admin privilege 15 secret timb@2024

line vty 0 4
login local
transport input ssh

line con 0
login local
```

## AAA + RADIUS Authentication

All device logins are authenticated via RADIUS server (192.168.200.12). This ensures centralized credential management for all network devices.

```
! For Routers and 2960 Switches:
aaa new-model
radius-server host 192.168.200.12 auth-port 1645 key timb@2024
aaa authentication login default group radius local
aaa authorization exec default group radius local

! For 3650 MLS Switches (new syntax):
radius server TIMB-RADIUS
address ipv4 192.168.200.12 auth-port 1645 acct-port 1646
key timb@2024
aaa authentication login default group radius local
```

## Banner MOTD

All devices display a legal warning banner upon access:

```
banner motd #
*********************************************
* TIMB BANK NETWORK DEVICE *
* AUTHORIZED ACCESS ONLY! *
* Unauthorized access is prohibited *
* and will be prosecuted to the *
* full extent of the law. *
* All activities are monitored and logged.*
* Disconnect IMMEDIATELY if you *
* are not an authorized user! *
*********************************************
#
```

## Security Features Summary

| Feature | Implementation | Coverage |
| --- | --- | --- |
| SSH v2 | All routers and switches | 100% of devices |
| AAA/RADIUS | Centralized auth server | All management access |
| ASA Firewall | 3-zone security model | Internet perimeter |
| DMZ | Isolated server zone | Public-facing servers |
| Guest VLAN | ACL restriction | VLAN 90 all sites |
| Banner MOTD | Legal warning | 100% of devices |
| NAT/PAT | IP hiding | All internal networks |
| VLAN Segmentation | Department isolation | All sites |

# 13. NETWORK MANAGEMENT

## NTP Configuration

All devices synchronize time with HQ NTP Server (192.168.200.13).

```
! On all routers and switches:
ntp server 192.168.200.13

! On NTP Server:
NTP Service → ON
```

## Syslog Logging

All devices send event logs to HQ Syslog Server (192.168.200.14).

```
! On all routers and switches:
logging host 192.168.200.14
logging on
```

## FTP Configuration Backup

All device configurations are backed up to HQ Backup Server (192.168.200.16) via FTP.

```
! Set FTP credentials on device:
ip ftp username admin
ip ftp password timb@2024

! Backup running config:
copy running-config ftp:
Address: 192.168.200.16
Filename: HQ-Router-backup.cfg
```

| Backup File | Device | Server |
|---|---|---|
| HQ-Router-backup.cfg | HQ-Router | 192.168.200.16 |
| HQ-MLS-backup.cfg | HQ-MLS | 192.168.200.16 |
| B1-Router-backup.cfg | B1-Router | 192.168.200.16 |
| B2-Router-backup.cfg | B2-Router | 192.168.200.16 |
| B3-Router-backup.cfg | B3-Router | 192.168.200.16 |
| B1-MLS-backup.cfg | B1-MLS | 192.168.200.16 |
| B2-MLS-backup.cfg | B2-MLS | 192.168.200.16 |
| B3-MLS-backup.cfg | B3-MLS | 192.168.200.16 |

# 14. SERVICES

## Email System

Internal email connects all HQ and branch staff using HQ Email Server (192.168.200.17) with SMTP and POP3 protocols.

| Account | Email Address | Server |
|---|---|---|
| HQ Admin | hq@timb.com | 192.168.200.17 |
| Branch 1 Staff | branch1@timb.com | 192.168.200.17 |
| Branch 2 Staff | branch2@timb.com | 192.168.200.17 |
| Branch 3 Staff | branch3@timb.com | 192.168.200.17 |

## Web Services — DMZ

| Service | URL | IP | Description |
|---|---|---|---|
| TIMB Bank Website | http://www.timb.com | 172.16.0.10 | Corporate website |
| Online Banking | http://banking.timb.com | 172.16.0.12 | Customer banking portal with MFA |
| Email Portal | http://mail.timb.com | 172.16.0.11 | Web email access |
| Google (Simulated) | http://www.google.com | 8.8.8.2 | Internet search simulation |

## DHCP Pools

DHCP Server (192.168.200.10) provides automatic IP assignment for all 10 VLANs at HQ. Each branch has its own DHCP server providing addresses for local VLANs.

# 15. BRANCH INDEPENDENT INTERNET

## Redundancy Design

Each branch has a dedicated internet link directly to the ISP Router — independent of HQ. If HQ internet fails, all branches remain online. This simulates SD-WAN redundancy behavior in Packet Tracer.

```
Traffic flow with HQ internet UP:
Branch PC → B-Router (Gi0/1) → ISP Vlan20x → Internet ■

Traffic flow with HQ internet DOWN:
Branch PC → B-Router (Gi0/1) → ISP Vlan20x → Internet ■
(Completely unaffected!)
```

## Branch Internet Links

| Branch | Router Interface | IP | ISP VLAN | ISP IP |
|---|---|---|---|---|
| Branch 1 | GigabitEthernet0/1 | 100.0.1.2/30 | VLAN 201 | 100.0.1.1 |
| Branch 2 | GigabitEthernet0/1 | 100.0.2.2/30 | VLAN 202 | 100.0.2.1 |
| Branch 3 | GigabitEthernet0/1 | 100.0.3.2/30 | VLAN 203 | 100.0.3.1 |

# 16. TROUBLESHOOTING & PROBLEMS SOLVED

## Major Problems Encountered and Solutions

### Problem: ASA blocking HTTP despite ping working...

| | |
|---|---|
| **Root Cause** | INSIDE_OUT ACL had 'deny ip any' before HTTP permit rules — ACLs processed top-down |
| **Solution** | Removed all ACLs — used ASA security levels which automatically allow inside→outside |

### Problem: Branch PCs not getting DHCP...

| | |
|---|---|
| **Root Cause** | ip helper-address not configured on MLS VLAN SVIs |
| **Solution** | Added ip helper-address 192.168.200.10 on all VLAN interfaces |

### Problem: OSPF not forming adjacency...

| | |
|---|---|
| **Root Cause** | Wrong network statements — HQ Router had 192.168.10.0 which belongs to MLS not HQ Router |
| **Solution** | Removed wrong network statement from HQ Router OSPF |

### Problem: GRE tunnel not working after adding branch internet...

| | |
|---|---|
| **Root Cause** | Two conflicting default routes — OSPF and static both pointing to different nexthops |
| **Solution** | Removed OSPF default-information originate on HQ Router, kept only direct ISP static routes |

### Problem: Crypto/IPSec commands invalid...

| | |
|---|---|
| **Root Cause** | Router IOS does not have security license in Packet Tracer |
| **Solution** | Used GRE tunnels instead — provided similar functionality |

### Problem: RADIUS login invalid...

| | |
|---|---|
| **Root Cause** | Client IP in RADIUS server was wrong — used ASA IP instead of router interface IP |
| **Solution** | Updated RADIUS server client IPs to match correct router interfaces |

### Problem: DMZ servers unreachable from branches...

| | |
|---|---|
| **Root Cause** | After removing OSPF default route, DMZ route was also lost at branches |
| **Solution** | Added static route: ip route 172.16.0.0 255.255.255.0 via WAN nexthop on each branch |

### Problem: Cloud1/DSL causing HTTP failures...

| | |
|---|---|
| **Root Cause** | Cloud1 port mapping was incorrect — Modem4 mapped to FastEthernet9 instead of Ethernet6 |
| **Solution** | Removed Cloud1 and DSL modem — connected ISP Router directly to ASA |

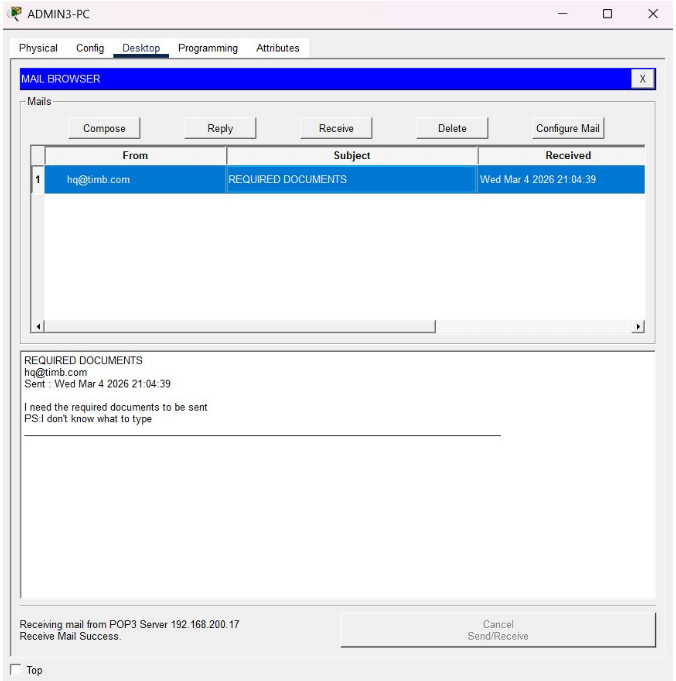
### Problem: 802.1X not working on 3650 MLS...

| | |
|---|---|
| **Root Cause** | Packet Tracer 3650 does not support dot1x port-control command |
| **Solution** | Applied AAA for management plane authentication instead — more practical approach |

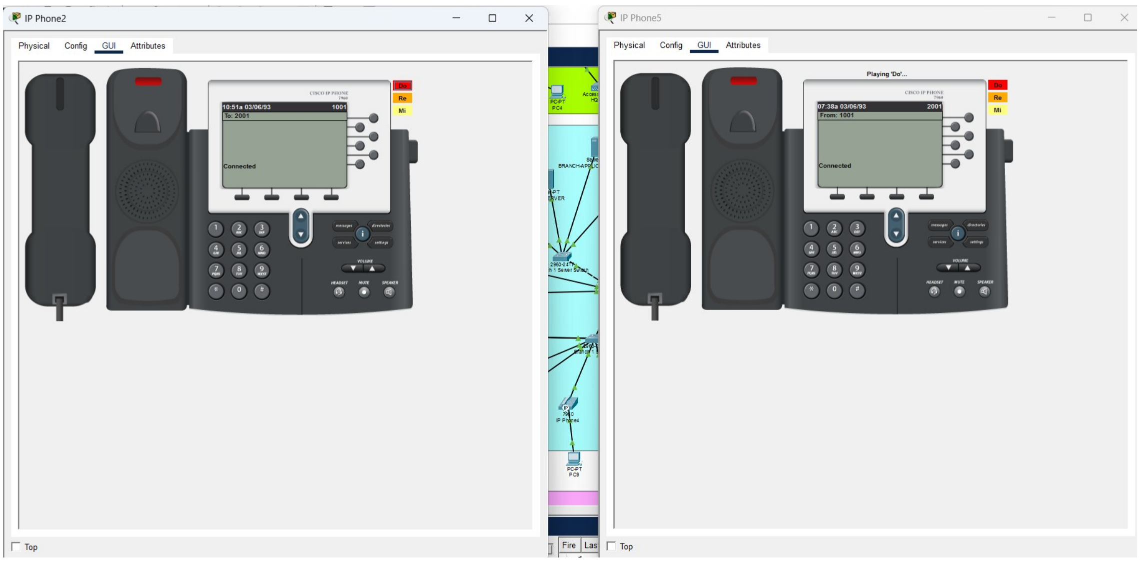
### Problem: ISP Router Fa1/x ports not accepting IP addresses...

| | |
|---|---|
| **Root Cause** | Fa1/x ports are Layer 2 switch ports on 2811 — cannot assign IPs directly |
| **Solution** | Created VLAN interfaces (Vlan201-203) and assigned ports to those VLANs |

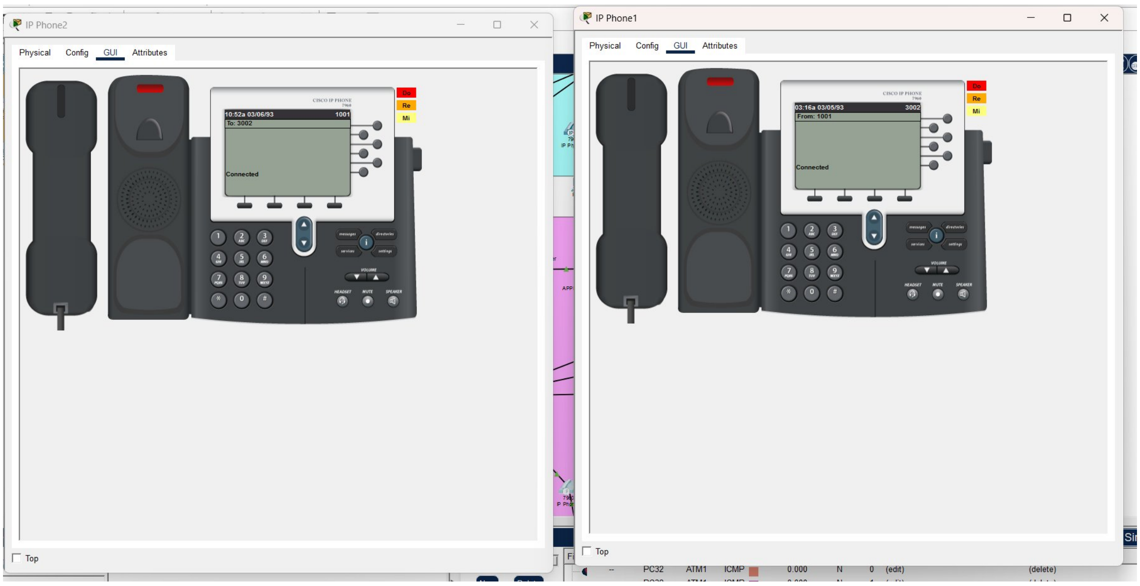
# 17. SCREENSHOTS & EVIDENCE



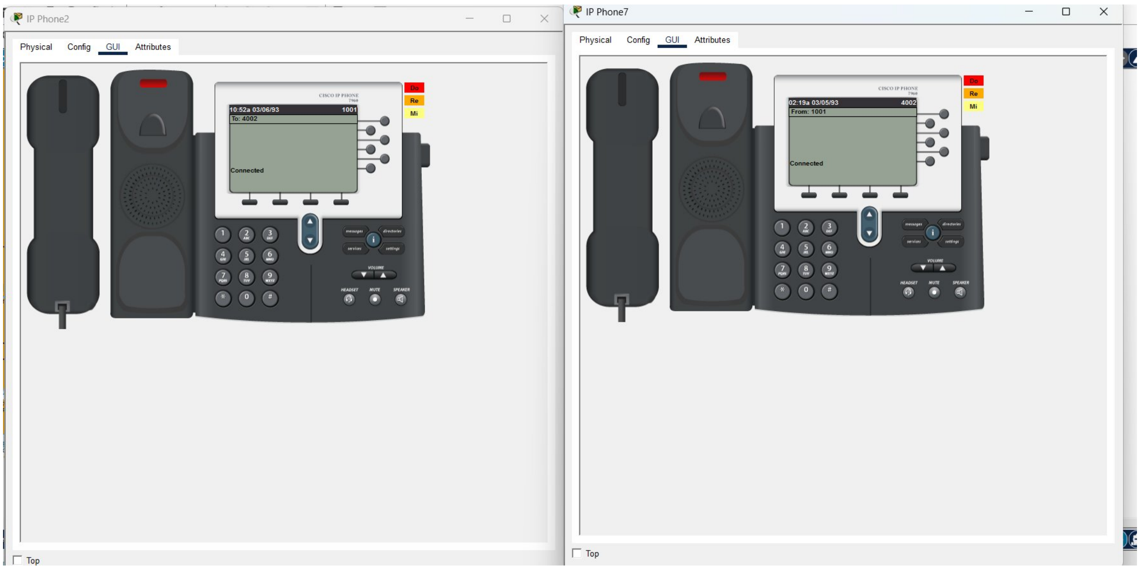

*Email received at Branch 3 from HQ — Internal email system working*



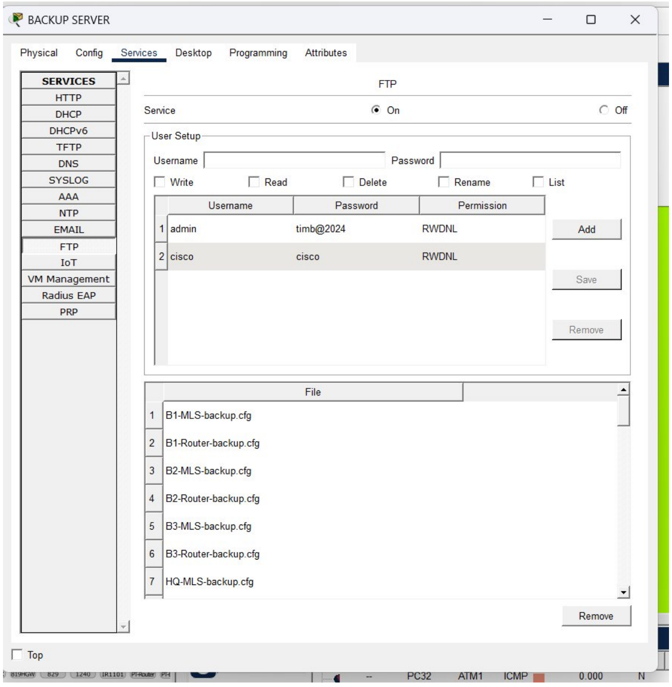

*VoIP Call — HQ Phone (1001) calling Branch 1 Phone (2001) — Connected*



*VoIP Call — HQ Phone (1001) calling Branch 2 Phone (3002) — Connected*



*VoIP Call — HQ Phone (1001) calling Branch 3 Phone (4002) — Connected*




*FTP Backup Server — All device configs backed up successfully*

```
ciscoasa>enable
Password:
Invalid password
Password:
Invalid password
Password: ciscoasa#
ciscoasa#show nat
Auto NAT Policies (Section 2)
1 (dmz) to (outside) source static BANKING-SERVER 203.0.113.12
    translate_hits = 0, untranslate_hits = 0
2 (inside) to (outside) source dynamic BRANCH1_NET interface
    translate_hits = 0, untranslate_hits = 0
3 (inside) to (outside) source dynamic BRANCH2_NET interface
    translate_hits = 0, untranslate_hits = 0
4 (inside) to (outside) source dynamic BRANCH3_NET interface
    translate_hits = 105, untranslate_hits = 144
5 (dmz) to (outside) source static EMAIL-SERVER 203.0.113.11
    translate_hits = 0, untranslate_hits = 0
6 (inside) to (outside) source dynamic HQ_NET interface
    translate_hits = 5, untranslate_hits = 3
7 (dmz) to (outside) source static WEB-SERVER 203.0.113.10
    translate_hits = 0, untranslate_hits = 0
```

*ASA show nat — NAT translations active for all networks*

```
ciscoasa#show xlate
18 in use, 18 most used
Flags: D - DNS, e - extended, I - identity, i - dynamic, r - portmap, s - static, T - twice, N -
net-to-net
NAT from dmz:172.16.0.12/32 to outside:203.0.113.12/32 flags s idle 01:28:14,  timeout 0:00:00
TCP-other PAT from inside:172.22.30.3/1026 to outside:203.0.113.2/1024 flags i idle 01:04:08,
timeout 0:00:30
TCP-other PAT from inside:172.22.30.3/1025 to outside:203.0.113.2/1025 flags i idle 01:05:06,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1026 to outside:203.0.113.2/1026 flags i idle 01:06:47,
timeout 0:00:30
TCP-other PAT from inside:172.22.80.4/1027 to outside:203.0.113.2/1027 flags i idle 00:36:30,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1028 to outside:203.0.113.2/1028 flags i idle 01:06:06,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1029 to outside:203.0.113.2/1029 flags i idle 01:06:02,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1030 to outside:203.0.113.2/1030 flags i idle 01:05:30,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1031 to outside:203.0.113.2/1031 flags i idle 01:04:21,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1032 to outside:203.0.113.2/1032 flags i idle 01:04:20,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1033 to outside:203.0.113.2/1033 flags i idle 01:04:19,
timeout 0:00:30
TCP-other PAT from inside:172.22.30.3/1027 to outside:203.0.113.2/1034 flags i idle 00:49:50,
timeout 0:00:30
TCP-other PAT from inside:172.22.30.3/1030 to outside:203.0.113.2/1035 flags i idle 00:47:16,
timeout 0:00:30
TCP-other PAT from inside:172.22.80.4/1026 to outside:203.0.113.2/1036 flags i idle 00:44:17,
timeout 0:00:30
TCP-other PAT from inside:172.22.10.3/1034 to outside:203.0.113.2/1037 flags i idle 00:42:47,
timeout 0:00:30
NAT from dmz:172.16.0.11/32 to outside:203.0.113.11/32 flags s idle 01:28:14,  timeout 0:00:00
TCP-other PAT from inside:192.168.30.3/1029 to outside:203.0.113.2/1038 flags i idle 00:11:54,
timeout 0:00:30
NAT from dmz:172.16.0.10/32 to outside:203.0.113.10/32 flags s idle 01:28:14,  timeout 0:00:00
```

*ASA show xlate — Active NAT translations from branches through HQ*

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l admin 192.168.1.2

Password:
% Login invalid


Password:
% Login invalid


Password:

[Connection to 192.168.1.2 closed by fo
C:\>ssh -l admin 192.168.1.2

Password:
% Login invalid


Password:
% Login invalid


Password:
```

*RADIUS AAA — SSH login authenticated via RADIUS server*

| | Time | HostName | Message |
|---|---|---|---|
| 1 | - | 10.0.0.2 | %SYS-5-CONFIG_I: Configure... |
| 2 | - | 10.0.0.2 | %SYS-6-LOGGINGHOST_STARTSTOP... |
| 3 | - | 172.20.0.2 | %SYS-5-CONFIG_I: Configure... |
| 4 | - | 172.20.0.2 | %SYS-6-LOGGINGHOST_STARTSTOP... |
| 5 | - | 10.0.1.2 | %SYS-5-CONFIG_I: Configure... |
| 6 | - | 10.0.1.2 | %SYS-6-LOGGINGHOST_STARTSTOP... |
| 7 | - | 172.21.0.2 | %SYS-5-CONFIG_I: Configure... |
| 8 | - | 172.21.0.2 | %SYS-6-LOGGINGHOST_STARTSTOP... |
| 9 | - | 10.0.2.2 | %SYS-5-CONFIG_I: Configure... |
| 10 | - | 10.0.2.2 | %SYS-6-LOGGINGHOST_STARTSTOP... |
| 11 | - | 172.22.0.2 | %SYS-5-CONFIG_I: Configure... |
| 12 | - | 172.22.0.2 | %SYS-6-LOGGINGHOST_STARTSTOP... |

*Syslog Server — Receiving logs from all branch routers and MLS devices*

Device Name: ISP-ROUTER
Custom Device Model: 2811 IOS15
Hostname: ISP_Router

| Port | Link | VLAN | IP Address | IPv6 Address | MAC Address |
|---|---|---|---|---|---|
| FastEthernet0/0 | Up | -- | 203.0.113.1/30 | <not set> | 0040.0B48.9A01 |
| FastEthernet0/1 | Down | -- | <not set> | <not set> | 0040.0B48.9A02 |
| FastEthernet1/0 | Up | 8 | -- | <not set> | 0001.4331.3C01 |
| FastEthernet1/1 | Down | 8 | -- | <not set> | 0001.4331.3C02 |
| FastEthernet1/2 | Down | 8 | -- | <not set> | 0001.4331.3C03 |
| FastEthernet1/3 | Up | 8 | -- | <not set> | 0001.4331.3C04 |
| FastEthernet1/4 | Down | 8 | -- | <not set> | 0001.4331.3C05 |
| FastEthernet1/5 | Down | 8 | -- | <not set> | 0001.4331.3C06 |
| FastEthernet1/6 | Down | 8 | -- | <not set> | 0001.4331.3C07 |
| FastEthernet1/7 | Down | 1 | -- | <not set> | 0001.4331.3C08 |
| FastEthernet1/8 | Down | 1 | -- | <not set> | 0001.4331.3C09 |
| FastEthernet1/9 | Down | 1 | -- | <not set> | 0001.4331.3C0A |
| FastEthernet1/10 | Down | 1 | -- | <not set> | 0001.4331.3C0B |
| FastEthernet1/11 | Down | 1 | -- | <not set> | 0001.4331.3C0C |
| FastEthernet1/12 | Down | 1 | -- | <not set> | 0001.4331.3C0D |
| FastEthernet1/13 | Down | 1 | -- | <not set> | 0001.4331.3C0E |
| FastEthernet1/14 | Down | 1 | -- | <not set> | 0001.4331.3C0F |
| FastEthernet1/15 | Down | 1 | -- | <not set> | 0001.4331.3C10 |
| Vlan1 | Down | 1 | <not set> | <not set> | 000C.85DB.382B |
| Vlan8 | Up | 8 | 8.8.8.1/24 | <not set> | 000C.85DB.3801 |

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > ISP-ROUTER

*ISP Router — Interface summary showing all connections*

Device Name: BRANCH1
Device Model: 2911
Hostname: B1-Router

| Port | Link | VLAN | IP Address | IPv6 Address | MAC Address |
|---|---|---|---|---|---|
| GigabitEthernet0/0 | Up | -- | 172.20.0.1/30 | <not set> | 000B.BEC4.8201 |
| GigabitEthernet0/1 | Down | -- | <not set> | <not set> | 000B.BEC4.8202 |
| GigabitEthernet0/2 | Down | -- | <not set> | <not set> | 000B.BEC4.8203 |
| Serial0/0/0 | Up | -- | <not set> | <not set> | <not set> |
| Serial0/0/0.1 | Up | -- | 10.0.0.2/30 | <not set> | <not set> |
| Serial0/0/0.2 | Up | -- | 10.1.0.1/30 | <not set> | <not set> |
| Serial0/0/0.3 | Up | -- | 10.2.0.1/30 | <not set> | <not set> |
| Serial0/0/1 | Down | -- | <not set> | <not set> | <not set> |
| Serial0/1/0 | Down | -- | <not set> | <not set> | <not set> |
| Serial0/1/1 | Down | -- | <not set> | <not set> | <not set> |
| Tunnel1 | Up | -- | 10.10.1.2/30 | <not set> | 0006.2A1A.743A |
| Vlan1 | Down | 1 | <not set> | <not set> | 0060.5C14.64B5 |

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > BRANCH1

*Branch 1 Router — Interface summary showing all active interfaces*

# 18. PROJECT SUMMARY

## Complete Feature List

| Category | Feature | Status |
|---|---|---|
| Infrastructure | HQ + 3 Branches configured | ■ Complete |
| Infrastructure | 10 VLANs per site | ■ Complete |
| Infrastructure | DHCP for all VLANs | ■ Complete |
| WAN | Frame Relay full mesh | ■ Complete |
| WAN | OSPF dynamic routing | ■ Complete |
| WAN | GRE VPN tunnels | ■ Complete |
| Internet | HQ internet via ASA | ■ Complete |
| Internet | Branch independent internet | ■ Complete |
| Internet | NAT for all networks | ■ Complete |
| Security | ASA Firewall (3 zones) | ■ Complete |
| Security | DMZ with 3 servers | ■ Complete |
| Security | SSH v2 all devices | ■ Complete |
| Security | AAA + RADIUS auth | ■ Complete |
| Security | Banner MOTD | ■ Complete |
| Security | Guest VLAN restriction | ■ Complete |
| Voice | VoIP local calls | ■ Complete |
| Voice | VoIP inter-site calling | ■ Complete |
| Services | TIMB Bank website | ■ Complete |
| Services | Online Banking portal | ■ Complete |
| Services | Internal email system | ■ Complete |
| Services | Google server simulation | ■ Complete |
| Management | NTP time sync | ■ Complete |
| Management | Syslog centralized logging | ■ Complete |
| Management | FTP config backup | ■ Complete |
| Redundancy | No single point of failure | ■ Complete |

## Production Recommendations

For a real-world deployment, the following upgrades are recommended:

| Current (Packet Tracer) | Production Recommendation |
|---|---|

| | |
|---|---|
| Frame Relay WAN | MPLS VPN with guaranteed SLA |
| GRE Tunnels | IPSec over GRE (encrypted tunnels) |
| Single ISP | SD-WAN (Cisco Viptela) with dual ISP |
| Basic RADIUS | Cisco ISE with MFA/802.1X |
| Static routing (branches) | BGP for ISP redundancy |
| Single ASA | Redundant ASA pair (Active/Standby) |
| No QoS | DSCP marking for VoIP prioritization |
| Manual backup | Automated scheduled config backup |

## Certification Level Coverage

This project demonstrates knowledge spanning multiple Cisco certification levels:

| Certification | Topics Covered in This Project |
|---|---|
| CCNA 200-301 | VLANs, OSPF, NAT, ACLs, SSH, DHCP, STP, Switching |
| CCNA Security | ASA Firewall, DMZ, AAA, RADIUS, SSH hardening |
| CCNA Voice | CME, VoIP, Dial-peers, ephone-dn configuration |
| CCNP Enterprise | Frame Relay, GRE, Advanced routing, WAN design |
| Network Design | Redundancy, segmentation, enterprise topology |

**TIMB Bank Network Project — Successfully Implemented**
Designed and built by Jessica Ugoh
Nigeria
Electronics and Computer Engineering
March 2026